

Laporan Uji Beban (Load Test) API — PUPR Manfaat IPAL

Tanggal: 16 Juni 2026 Alat uji: k6 v2.0.0 Disiapkan untuk: Klien / Stakeholder

Ringkasan Eksekutif

API diuji dengan **50 pengguna aktif bersamaan** menembak 6 endpoint utama secara terus-menerus selama ~51 detik (total **3.889 permintaan**).

✅ **Hasil: LULUS.** Seluruh endpoint merespons jauh di bawah target 500ms (p95 tertinggi hanya **33ms**), dengan **0% kegagalan**. Sistem proteksi rate-limit (anti-spam/DoS) juga terbukti aktif dan berfungsi.

Kesimpulan: API cepat, stabil, dan siap produksi.

Metodologi

Aspek	Keterangan
Skenario	50 virtual user, ramp bertahap (10 → 30 → 50), durasi ~51 detik
Beban	3.889 permintaan (~76 permintaan/detik)
Build	Produksi (node dist , logging produksi) — bukan mode pengembangan
Endpoint	6 endpoint baca utama (daftar, peta, pengguna, dsb.)
Target latency	p95 < 500ms (standar pengalaman pengguna)

Catatan: rate-limit dinaikkan sementara khusus saat pengukuran kapasitas, agar angka mencerminkan performa server sebenarnya (bukan ditahan oleh proteksi).

Hasil per Endpoint

Angka **p95** = 95% permintaan lebih cepat dari ini (indikator pengalaman terburuk yang wajar).

Endpoint	p95	Target p95	Status
GET /ipals/map	16ms	<500ms	✅ Lulus
GET /surveys/map	17ms	<500ms	✅ Lulus
GET /ipals	19ms	<500ms	✅ Lulus

Endpoint	p95	Target p95	Status
GET /rbac/roles	24ms	<500ms	✓ Lulus
GET /users	25ms	<500ms	✓ Lulus
GET /surveys	33ms	<500ms	✓ Lulus

Metrik keseluruhan:

Metrik	Nilai
Total permintaan	3.889
Tingkat kegagalan	0,00%
Throughput	~76 permintaan/detik
Latency median	7ms
Latency p95 (gabungan)	23ms

Temuan & Catatan

- Performa seragam & sehat.** Tidak ada endpoint yang menonjol lambat. Endpoint tercepat ~16ms (p95), terlambat ~33ms (p95) — semua jauh di bawah ambang 500ms.
- Tanpa kegagalan** pada 3.889 permintaan di bawah beban konkuren.
- Proteksi keamanan aktif:**
 - Rate-limit (throttle) membatasi 120 permintaan/menit per IP — mencegah spam & brute-force.
 - Endpoint login dibatasi lebih ketat (5/menit) — anti brute-force.
 - Security headers (Helmet) aktif pada seluruh response.

Rekomendasi

- API layak naik produksi dari sisi performa & keamanan dasar.
- Pemantauan berkelanjutan (mis. Grafana) disarankan untuk memantau tren di produksi.
- Uji ulang setelah perubahan besar pada query/skema data.

Lampiran

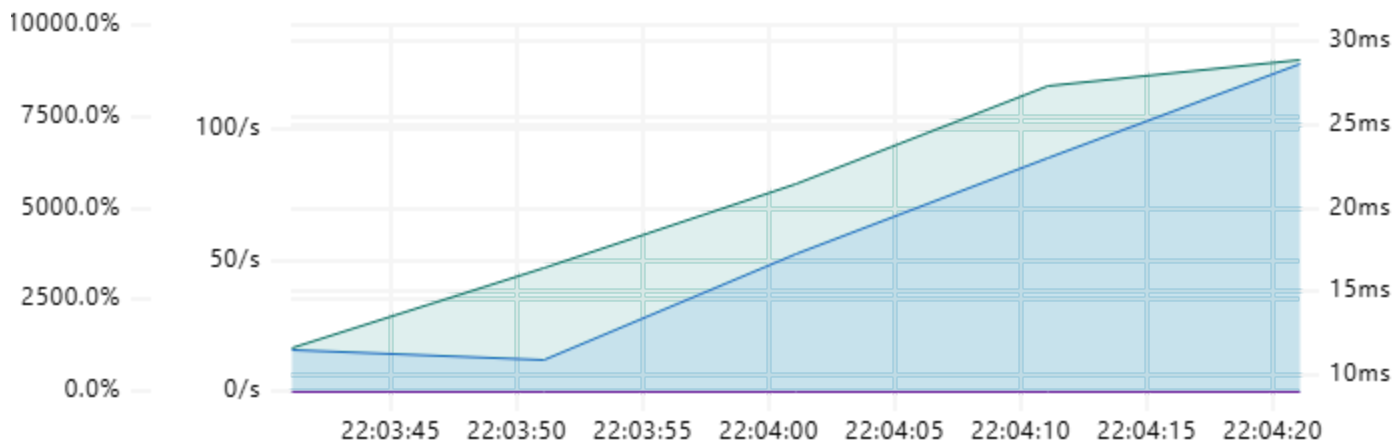
- Grafik interaktif:** `report.html` (buka di browser) — grafik latency, jumlah pengguna, dan throughput sepanjang waktu uji.
- Script uji:** `test/load/api-load.js` (dapat dijalankan ulang kapan saja).



Overview

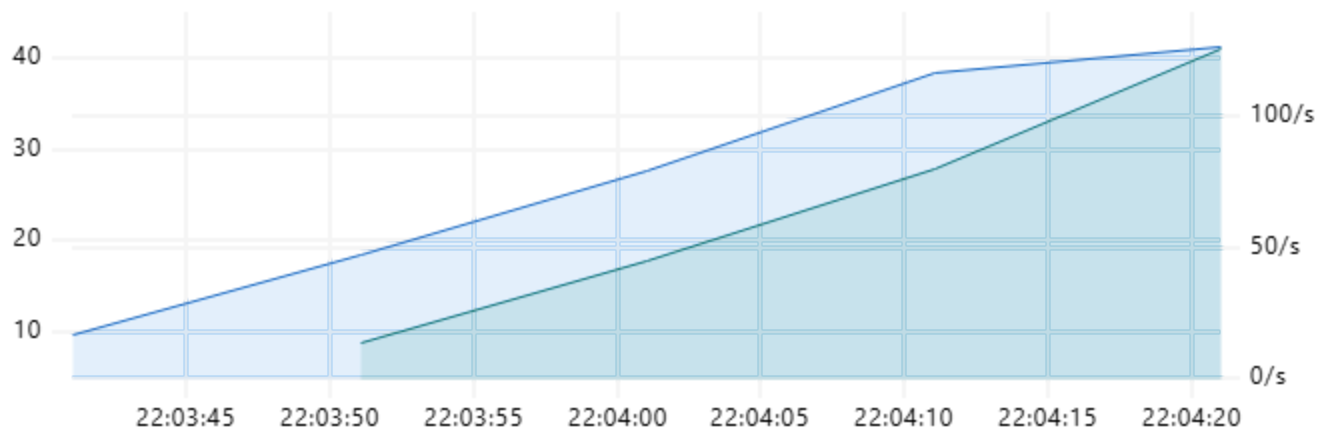
This chapter provides an overview of the most important metrics of the test run. Graphs plot the value of metrics over time.

HTTP Performance overview



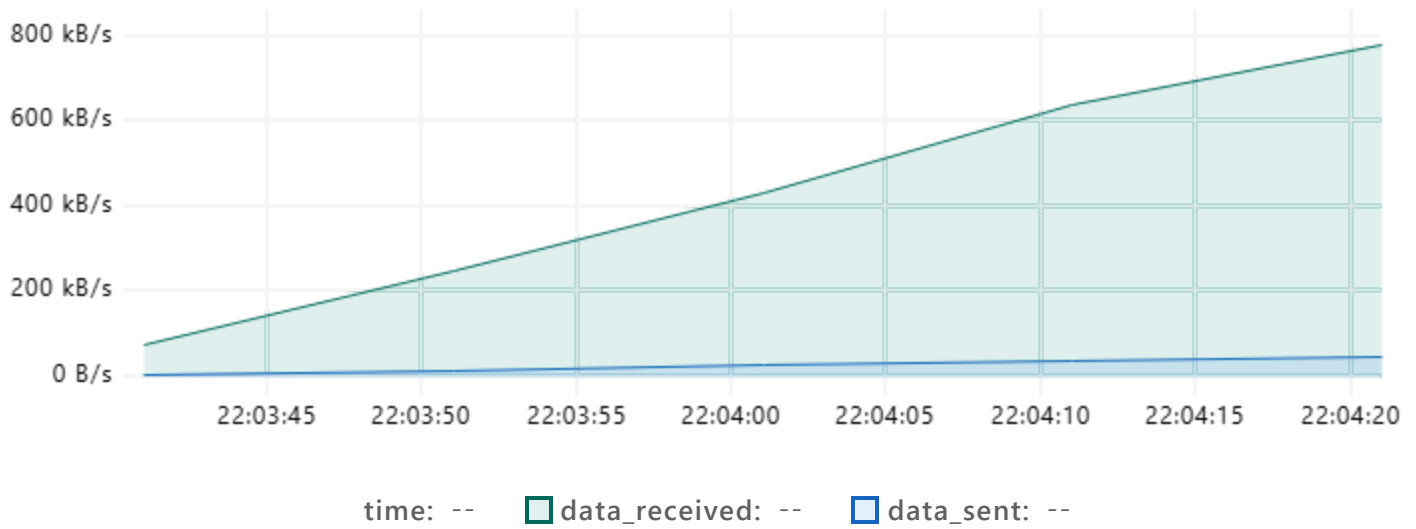
time: -- Request Rate: -- Request Duration p(95): -- Request Failed: --

VUs

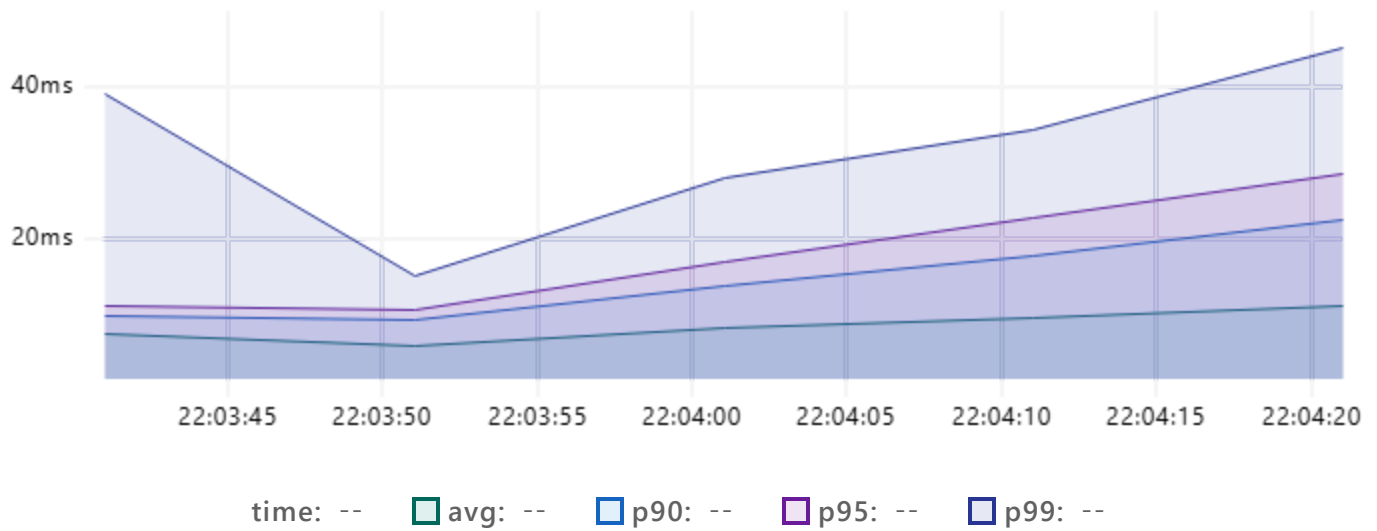


time: -- vus: -- http_reqs: --

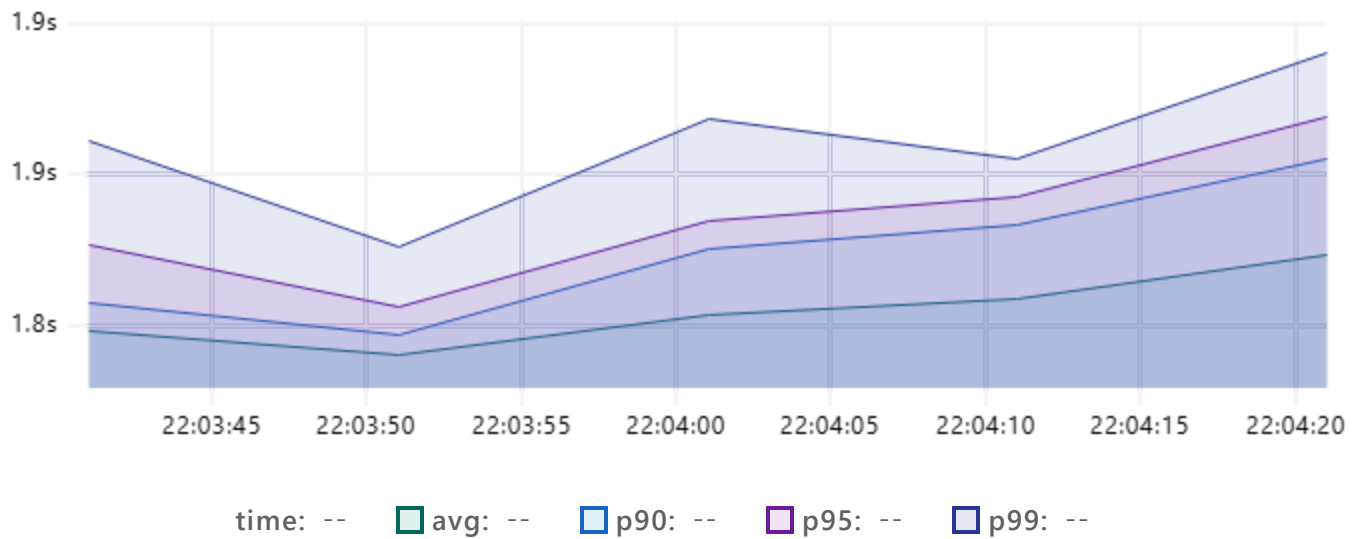
Transfer Rate



HTTP Request Duration



Iteration Duration



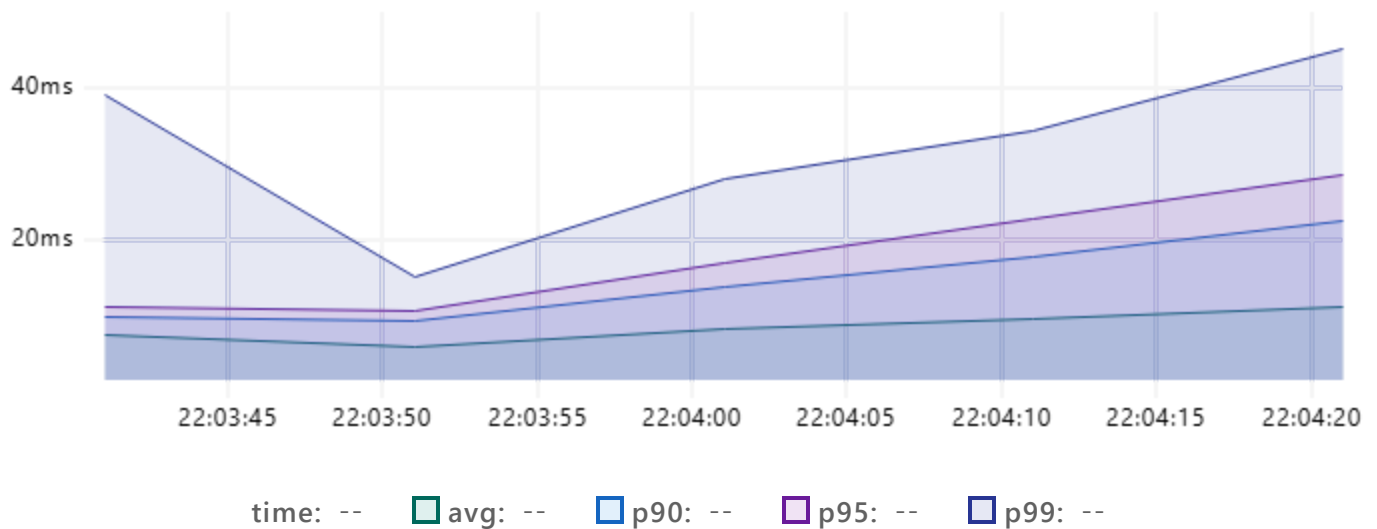
Timings

This chapter provides an overview of test run HTTP timing metrics. Graphs plot the value of metrics over time.

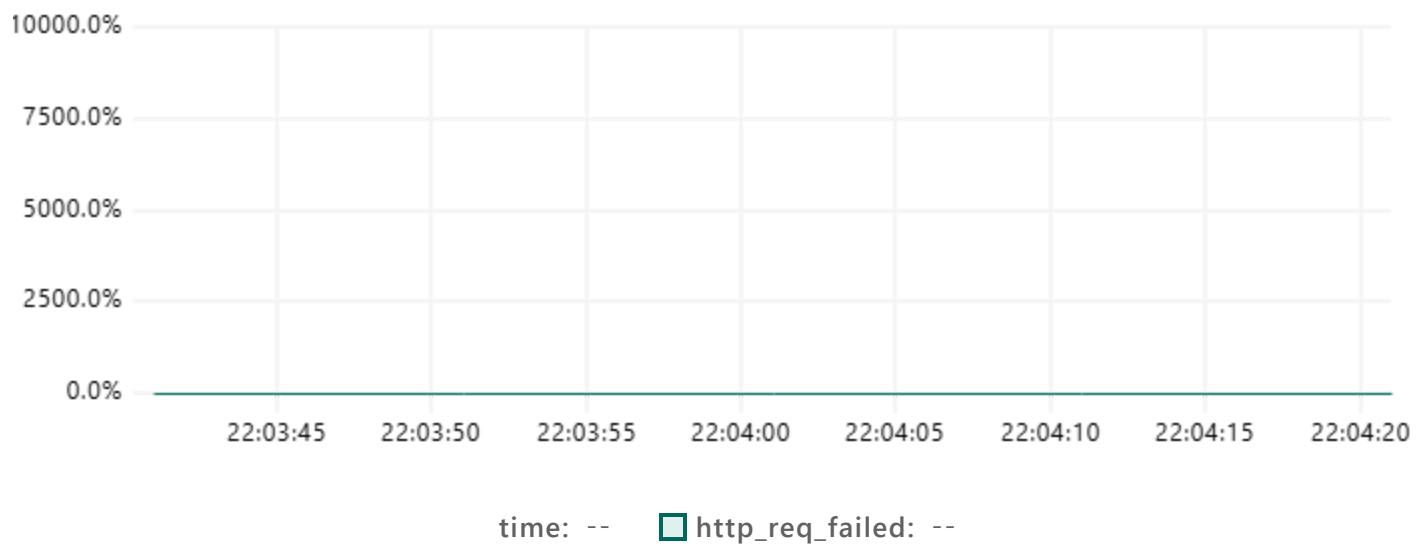
● HTTP

These metrics are generated only when the test makes HTTP requests.

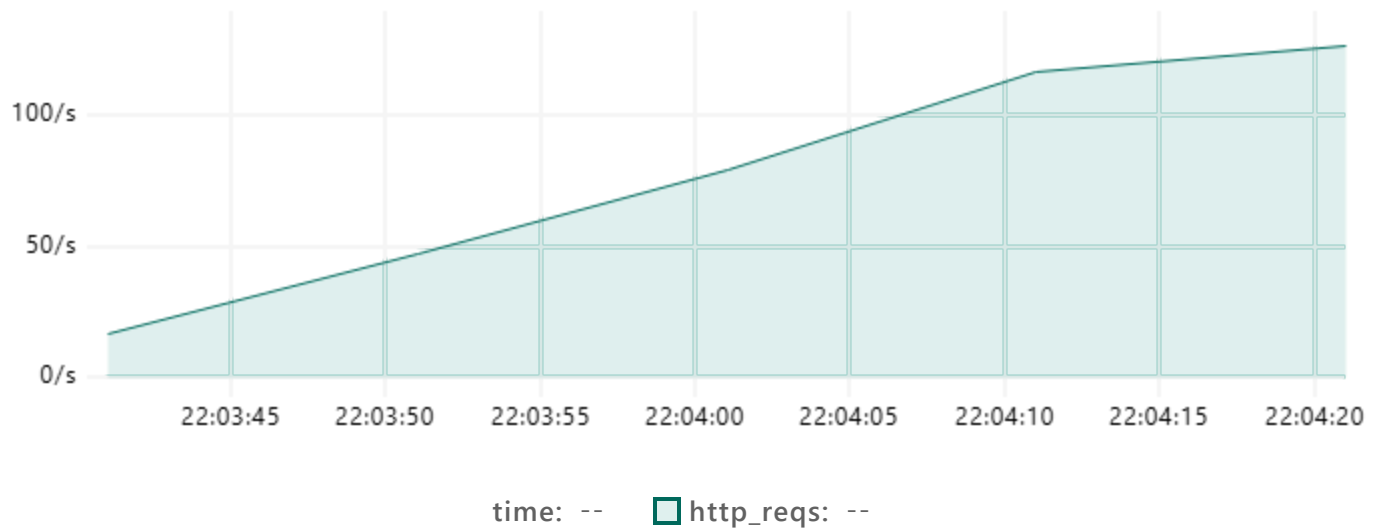
Request Duration



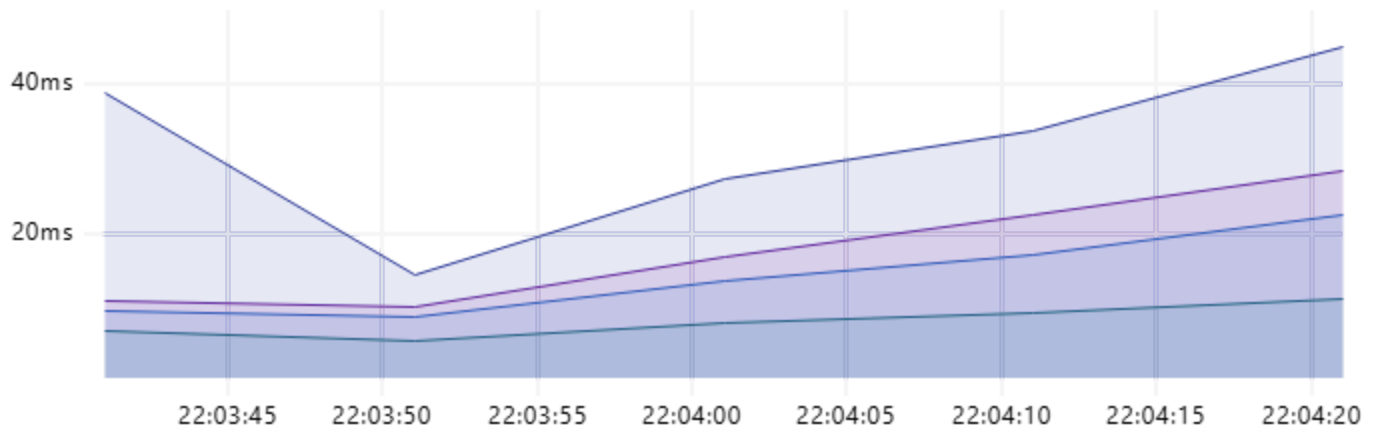
Request Failed Rate



Request Rate

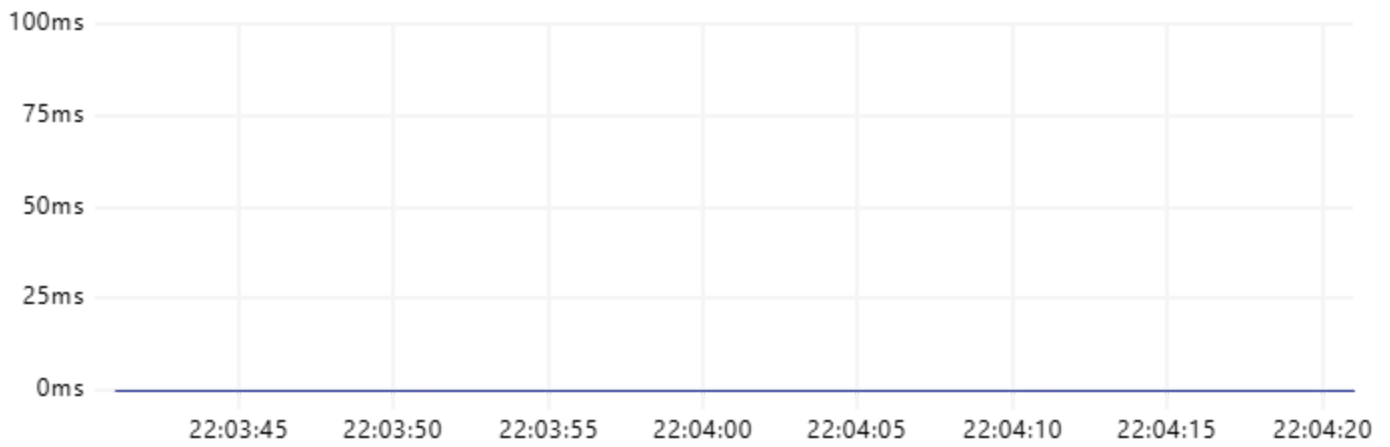


Request Waiting



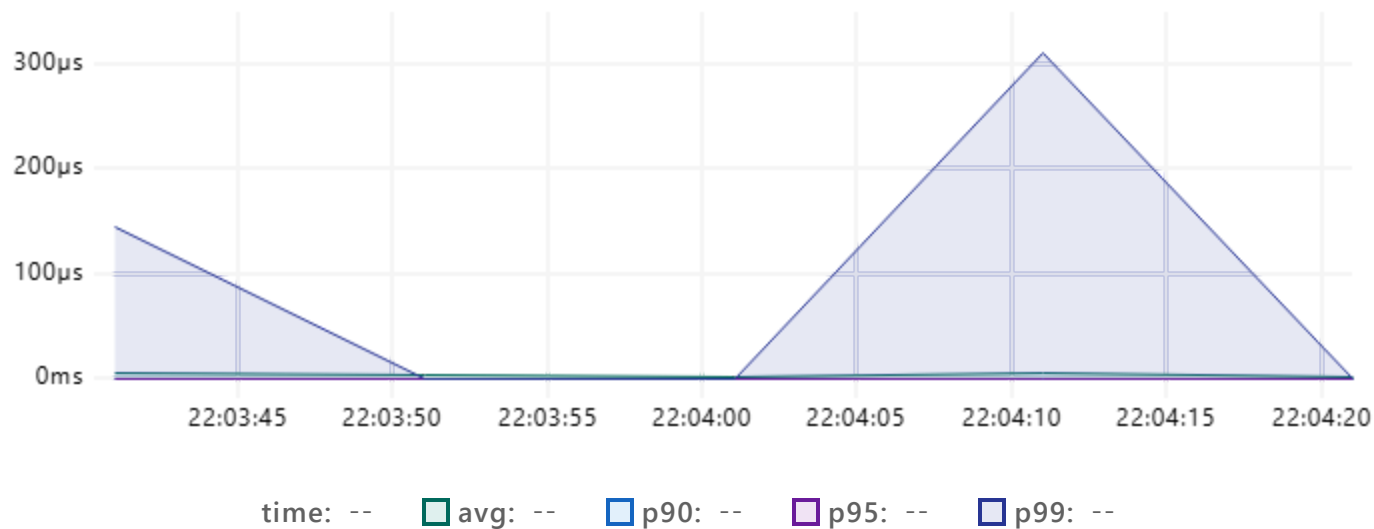
time: -- avg: -- p90: -- p95: -- p99: --

TLS handshaking

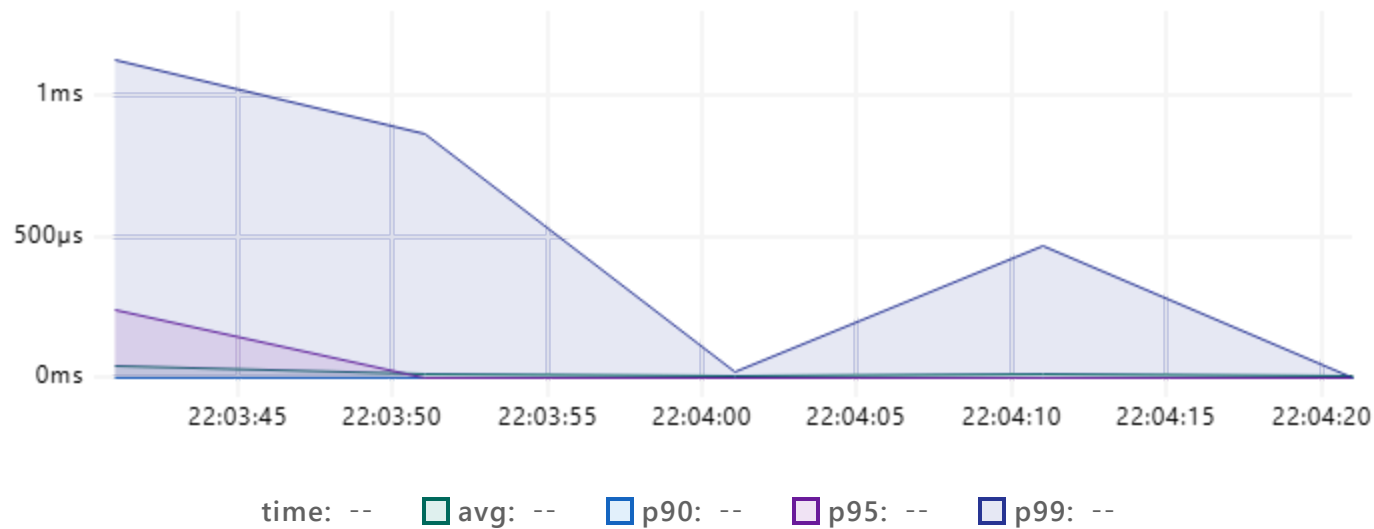


time: -- avg: -- p90: -- p95: -- p99: --

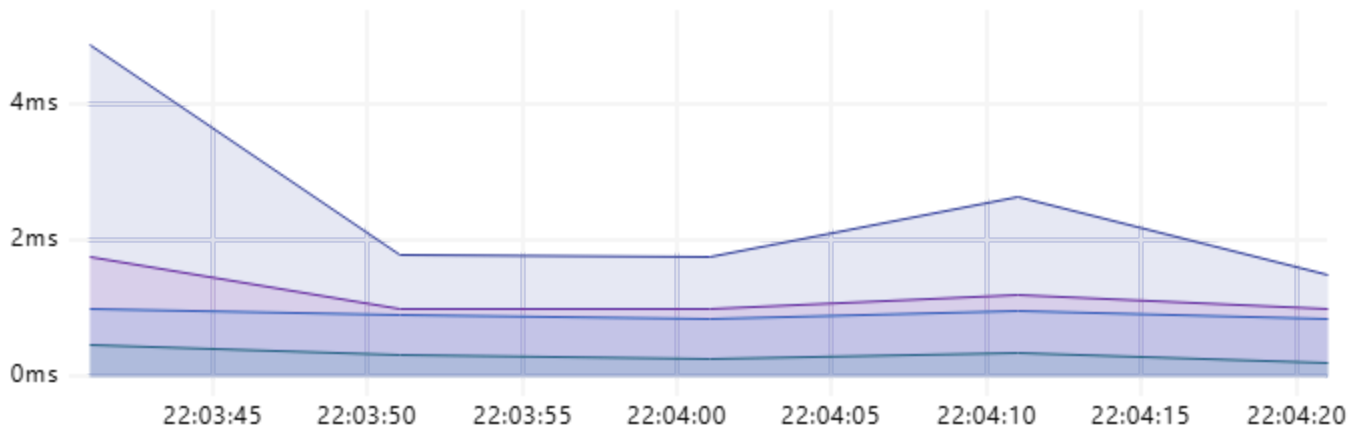
Request Sending



Request Connecting

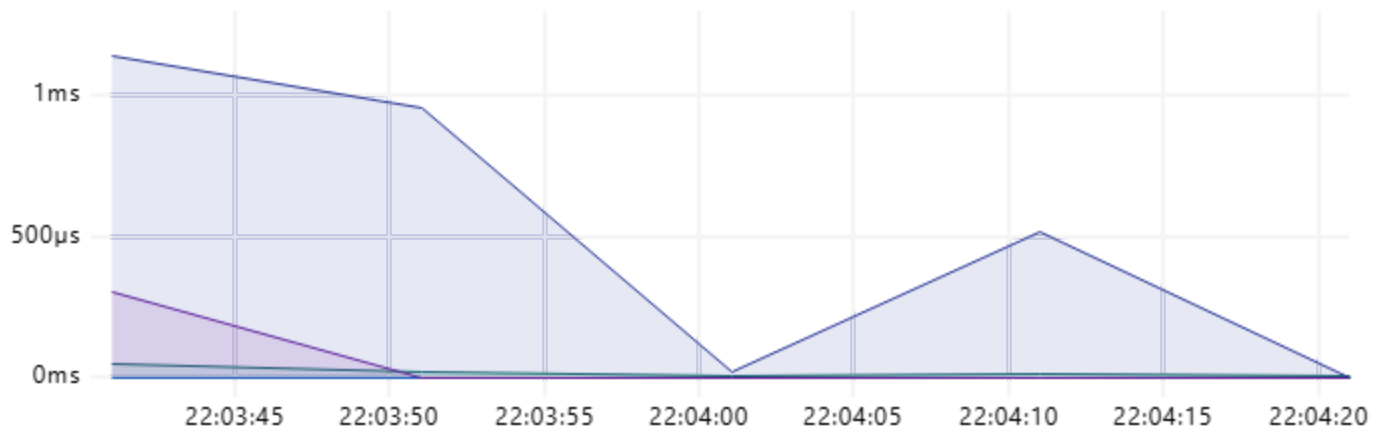


Request Receiving



time: -- avg: -- p90: -- p95: -- p99: --

Request Blocked



time: -- avg: -- p90: -- p95: -- p99: --

Summary

This chapter provides a summary of the test run metrics. The tables contains the aggregated values of the metrics for the entire test run.

Trends

metric	avg	max	med	min	p90	p95	p99
http_req_blocked	12µs	4ms	0ms	0ms	0ms	0ms	520µs
http_req_connecting	11µs	3ms	0ms	0ms	0ms	0ms	506µs

metric	avg	max	med	min	p90	p95	p99
http_req_duration	9ms	87ms	7ms	422µs	17ms	23ms	38ms
http_req_receiving	293µs	6ms	0ms	0ms	916µs	1ms	2ms
http_req_sending	3µs	1ms	0ms	0ms	0ms	0ms	0ms
http_req_tls_handshaking	0ms	0ms	0ms	0ms	0ms	0ms	0ms
http_req_waiting	9ms	87ms	7ms	422µs	17ms	22ms	38ms
iteration_duration	1s	1s	1s	1s	1s	1s	1s

Counters

metric	count	rate
data_received	22.2 MB	432 kB/s
data_sent	1.36 MB	26.6 kB/s
http_reqs	3.9k	75.84/s
iterations	648	12.64/s

Rates

metric	rate
checks	100.0%
http_req_failed	0.0%

Gauges

metric	value
vus	2
vus_max	50

Select a time interval by holding down the mouse on any graph to zoom. To cancel zoom, double click on any graph.